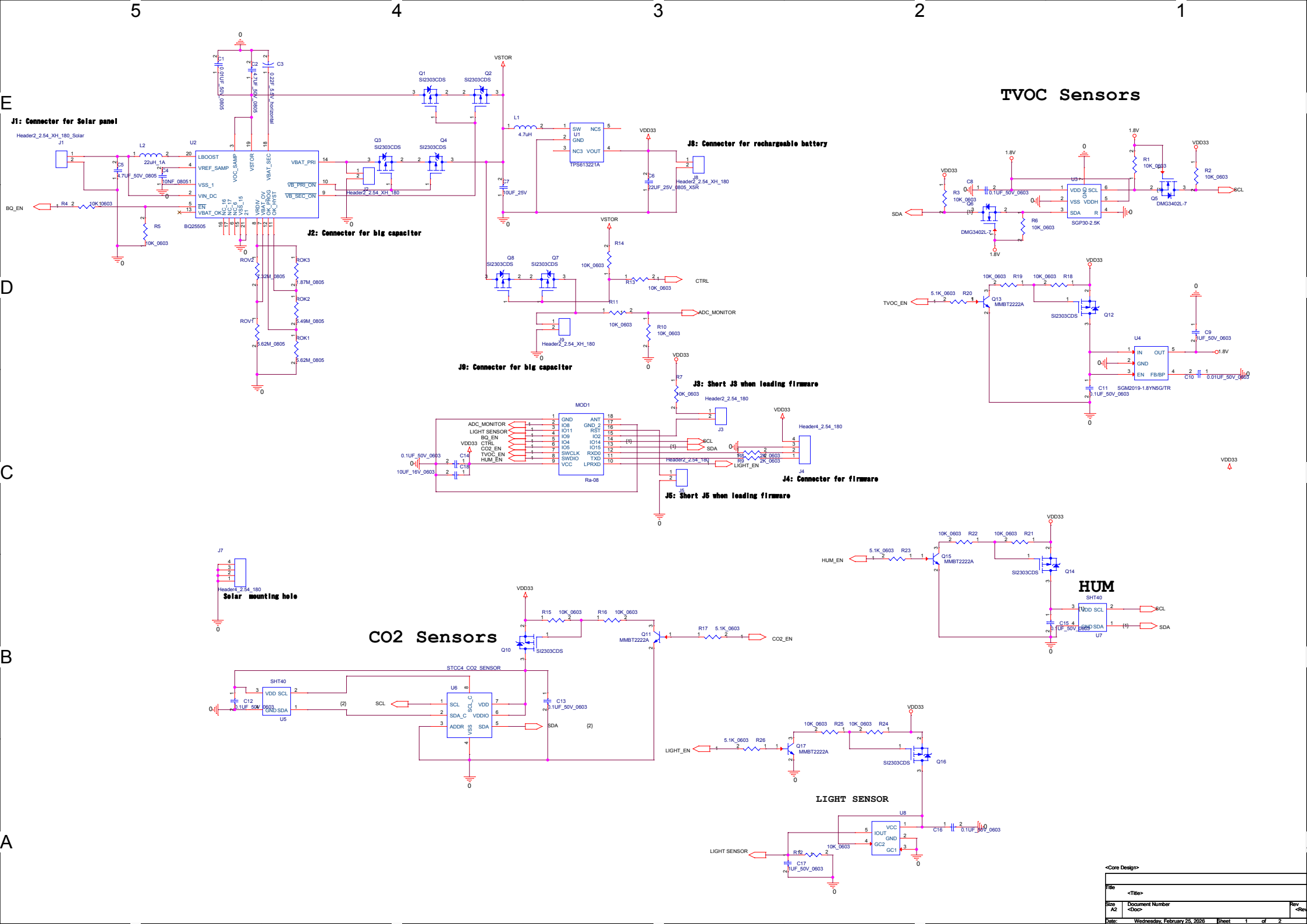
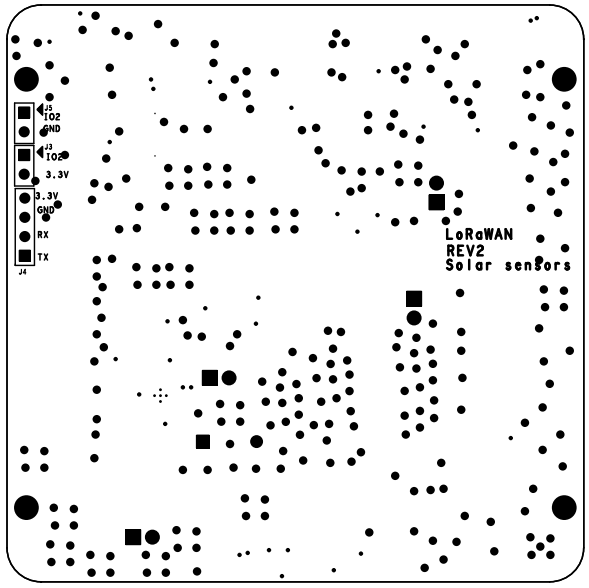
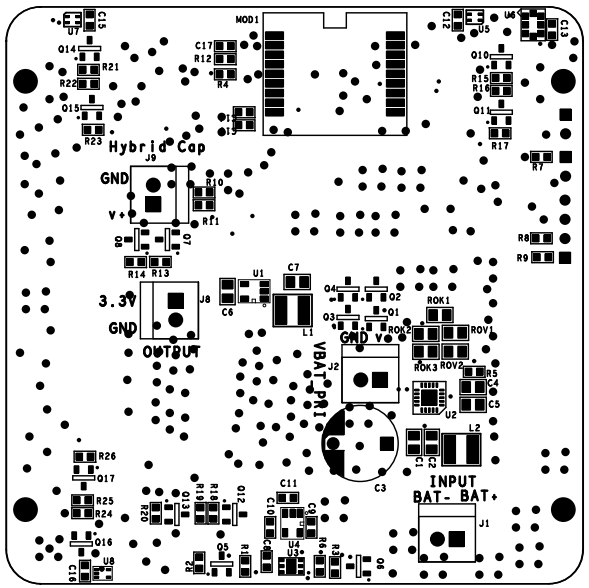

5
4
3
2
1
E
D
C
B
A
J1: Connector for Solar panel
Header2_2.54_XH_180_Solar
J1
L2
22uH_1A
U2
LBOOST
VREF_SAMP
VSS_1
VIN_DC
EN
VBAT_OK
VOC_SAMP
VSTOR
VBAT_SEC
VBAT_PRI
VB_PRI_ON
VB_SEC_ON
VRDIV
VBAT_OV
VBAT_OK_HYST
OK_HYST
BQ25505
BQ_EN
R4 10K1603
R5 10K_0603
C5 4.7UF_50V_0805
C4 10NF_08051
C1 0.01UF_50V_0805
C2 4.7UF_50V_0805
C3 0.22F_5.5V_horizontal
Q1 SI2303CDS
Q2 SI2303CDS
Q3 SI2303CDS
Q4 SI2303CDS
VSTOR
L1 4.7uH
U1 TPS613221A
SW
NC5
GND
NC3
VOUT
VDD33
C7 10UF_25V
C6 22UF_25V_0805_X5R
J8: Connector for rechargeable battery
J8 Header2_2.54_XH_180
J2: Connector for big capacitor
Header2_2.54_XH_180
ROV2
ROV1
ROK3 1.87M_0805
ROK2 5.49M_0805
ROK1 5.62M_0805
5.32M_0805
5.62M_0805
Q8 SI2303CDS
Q7 SI2303CDS
R14 10K_0603
R13 10K_0603
CTRL
R11 10K_0603
R10 10K_0603
ADC_MONITOR
J9 Header2_2.54_XH_180
J9: Connector for big capacitor
R7 10K_0603
J3: Short J3 when loading firmware
Header2_2.54_180
J3
MOD1
Ra-08
ADC_MONITOR
LIGHT SENSOR
BQ_EN
CTRL
CO2_EN
TVOC_EN
HUM_EN
SCL
SDA
C14 0.1UF_50V_0603
C18 10UF_16V_0603
R8 2K_0603
R9 2K_0603
LIGHT_EN
Header4_2.54_180
J4
J4: Connector for firmware
J5
J5: Short J5 when loading firmware
J7
Header4_2.54_180
Solar mounting hole
TVOC Sensors
U3 SGP30-2.5K
R1 10K_0603
R2 10K_0603
R3 10K_0603
C8 0.1UF_50V_0603
Q5 DMG3402L-7
Q6 DMG3402L-7
R6 10K_0603
1.8V
SCL
SDA
R19 10K_0603
R18 10K_0603
R20 5.1K_0603
Q13 MMBT2222A
Q12 SI2303CDS
TVOC_EN
U4 SGM2019-1.8YN5G/TR
C9 1UF_50V_0603
C10 0.01UF_50V_0603
C11 0.1UF_50V_0603
CO2 Sensors
Q10 SI2303CDS
R15 10K_0603
R16 10K_0603
Q11 MMBT2222A
R17 5.1K_0603
CO2_EN
STCC4_CO2_SENSOR
U6
SHT40
U5
C12 0.1UF_50V_0603
C13 0.1UF_50V_0603
SCL
SDA
HUM
R22 10K_0603
R21 10K_0603
R23 5.1K_0603
Q15 MMBT2222A
Q14 SI2303CDS
HUM_EN
U7
C15 0.1UF_50V_0603
LIGHT SENSOR
R25 10K_0603
R24 10K_0603
R26 5.1K_0603
Q17 MMBT2222A
Q16 SI2303CDS
LIGHT_EN
U8
C16 0.1UF_50V_0603
C17 1UF_50V_0603
R12 10K_0603
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Title
<Title>
Size A2
Document Number <Doc>
Rev <Rev>
Date: Wednesday, February 25, 2026
Sheet 1 of 2

<Core Design>

Title	<Title>
Size: B	Document Number: <Doc>
Date:	Monday, February 04, 2019
Sheet 2 of 3	


J5
IO2
GND
J3
IO2
3.3V
3.3V
GND
RX
TX
J4
LoRaWAN
REV2
Solar sensors


MOD1
Hybrid Cap
J9
GND
V+
J8
3.3V
GND
OUTPUT
J2
GND V
VBAT_PRI
INPUT
BAT- BAT+
J1
ROK1
ROV1
ROK2
ROV2
ROK3
U1
U2
U3
U4
U5
U6
U7
U8
L1
L2
C1
C2
C3
C5
C6
C7
C10
C11
C12
C13
C16
C17
R1
R2
R3
R4
R5
R7
R8
R9
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R14
R15
R16
R17
R18
R19
R20
R21
R22
R23
R24
R25
R26
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Q8
Q9
Q10
Q11
Q12
Q13
Q14
Q15
Q16
Q17
D1
D2
D3
D4